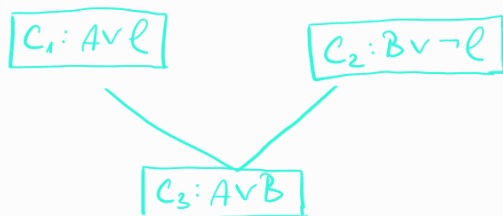


$$C_1 = A \vee \ell \quad \text{Rez} \quad A \vee B = C_3$$

$$C_2 = B \vee \neg \ell$$



$$C_1 = A \vee \ell$$

$$C_2 = B \vee \neg \ell$$

$$C_3 = \text{Rez}_2(C_1, C_2) = A \vee B$$

$$\frac{A \quad A \rightarrow B}{\vdash_{\text{mp}} B}$$

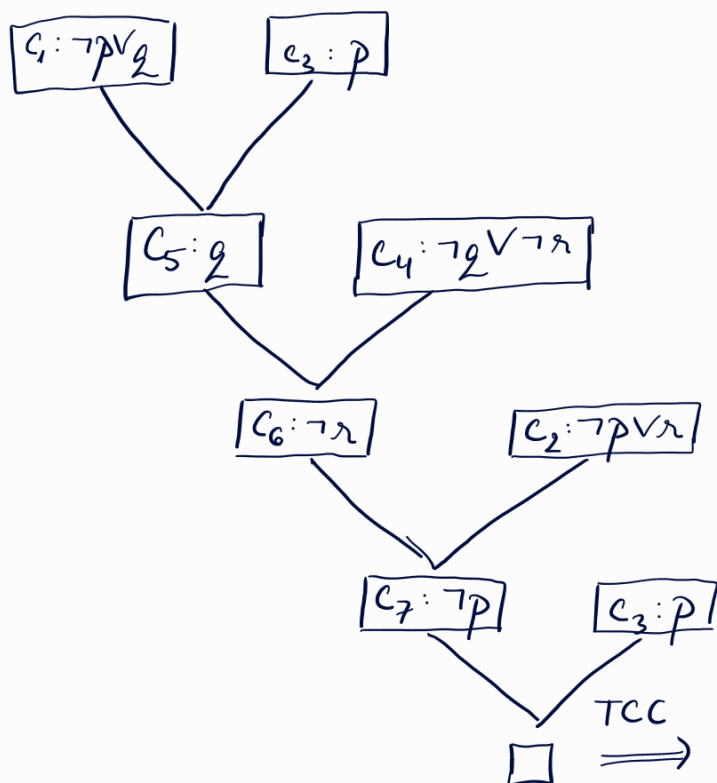
$$\frac{A \rightarrow B \quad \neg B}{\vdash_{\text{mt}} \neg A}$$

$$\frac{u \rightarrow v \quad v \rightarrow z}{\vdash_{\text{sil}} u \rightarrow z}$$

Teoreme
 u este tautologie $\Leftrightarrow \text{FNC}(\neg u) \vdash_{\text{Res}} \square$

Ex.:

$$S = \{\neg p \vee q, \neg p \vee r, p, \neg q \vee \neg r\}$$



DE CE TI, TABLETA
INVERSE

TCC = Teorema de corectitudine și completitudine

Problema

p = Ion face laboratorul
 q = Vasile face laboratorul
 r = Gheorghe face laboratorul

$$u_1: p \rightarrow (p \vee q)$$

$$u_2: p \rightarrow q$$

$$u_3: \neg q \rightarrow r$$

$$u_4: \neg r \rightarrow \neg p \vee q$$

$$u_5: \neg r$$

$$v: q$$

$$u_1: p \rightarrow (p \vee q) \equiv \neg p \vee p \vee q = c_1$$

$$u_2: p \rightarrow q \equiv \neg p \vee q = c_2$$

$$u_3: \neg q \rightarrow r \equiv q \vee r = c_3$$

$$u_4: \neg r \rightarrow \neg p \vee q \equiv r \vee \neg p \vee q = c_4$$

$$u_5: \neg r = c_5$$

$$v: q \Rightarrow \neg v \equiv \neg q = c_6$$

$$S_1 = \{\neg p \vee p \vee q, \neg p \vee q, q \vee r, r \vee \neg p \vee q, \neg r\}$$

Pas 1: eliminăm elementele tautologice

$$S_1 = \{\neg p \vee q, q \vee r, r \vee \neg p \vee q, \neg r\}$$

Pas 2:

$r \vee \neg p \vee q$ subsumată de $q \vee r$

$$S_1 = \{\neg p \vee q, q \vee r, \neg r\}$$

$$S_2 = \{\neg p \vee p \vee q, \neg p \vee q, q \vee r, r \vee \neg p \vee q, \neg r, \neg q\}$$

$$\Rightarrow S_2 = \{\neg p \vee q, q \vee r, \neg r, \neg q\}$$

Pas 3:

q -literal pur

$$S_1 = \{\neg r\}$$

$\neg r$ -literal pur

$$S_1 = \{\} = \emptyset \quad \text{-multime consistentă} \Rightarrow \text{pact necontradictoriu}$$

Pas 3(S_2):

$$\neg p \text{-literal pur} \Rightarrow S_2 = \{q, \neg r, \neg q\}$$

$$q \text{-clauză unitate} \Rightarrow S_2 = \{\neg r, \square\}$$

$$\neg r \text{-literal pur} \Rightarrow S_2 = \{\square\}$$

EU ZIL CA S-O

IMBATAT TOTI 3

inconsistentă $\Rightarrow$ Vasile și-a făcut laboratorul

$\emptyset$ (consistentă) sau $\square$ (inconsistentă)

Strategia mulțimii suport

$$S_2 = \{ \overset{C_1}{\neg p \vee p \vee q}, \overset{C_2}{\neg p \vee q}, \overset{C_3}{q \vee r}, \overset{C_4}{r \vee \neg p \vee q}, \overset{C_5}{\neg r}, \overset{C_6}{\neg q} \}$$

S_1 - e consistentă (din ipoteze)

Nu vom rezolva $C_1 \rightarrow C_5$ între ele

$$C_7 = \text{Rez}_q(C_6, C_2) = \neg p$$

$$C_8 = \text{Rez}_p(C_7, C_1)$$

$$C_3 = \text{Rez}_q(C_6, C_4) = r \vee \neg p$$

$$C_9 = \text{Rez}_q(C_6, C_3) = r$$

$$C_{10} = \text{Rez}_r(C_9, C_5) = \square$$

TCC

$\Rightarrow S_2$ e inconsistentă